

DARES Electronics: BMS and Power Systems

Mid-semester Break Plan

Prepared by (Alfred) Zi Wang

September 2026

1 System Overview

The power system is responsible for safely storing, distributing, monitoring, and supplying electrical energy to every subsystem on the drone.

High-Level Drone Power System



Figure 1: High level block diagram

with additional regulated outputs supplying:

Flight Controller, Sensors, Communications, Navigation Electronics, and Other Low-Power Loads.

2 Our Main Design Questions

- **What battery should we use?**

We need to determine the required battery voltage, capacity, discharge rate, mass, and chemistry based on the propulsion system, flight controller and expected flight time. Also check the input constraints (voltage, current) of the flight controller or power distribution board.

- **How much current does the drone require?**

The motor and ESC will dominate the power consumption. We need to estimate both continuous current and short-duration peak current.

- **How do we safely monitor the battery?**

The system should measure important quantities such as:

- Battery voltage
- Individual cell voltages
- Battery current
- Battery temperature
- State of Charge (SOC)

(Percentage of usable battery capacity remaining, e.g. 100%, 50%)

- **What protection do we need?**

Possible protection functions include:

- Over-voltage protection
- Under-voltage protection
- Over-current protection

- Short-circuit protection
- Over-temperature protection
- Cell balancing

- **How should power be distributed?**

We need to determine how the battery connects to:

- ESCs and motors
- Flight controller
- Sensors
- Radio / telemetry
- Navigation electronics
- Payload electronics

3 General Design Process

- Start by determining the expected electrical requirements of each subsystem.
- Create a **power budget** containing:
 - Operating voltage
 - Average current
 - Peak current
 - Average power
 - Maximum power
- Use propulsion data to estimate the maximum battery current.
- Select a battery voltage and capacity that satisfy the propulsion requirements and expected flight duration.
- Determine suitable connectors, cables, PCB traces, fuses, switches, and power distribution hardware based on the expected maximum current.
- Design the low-voltage power supplies required by the electronics.
- Implement battery manage system for voltage, current, and temperature monitoring, protection and, if required, cell balancing.
- Test each subsystem individually before integrating the complete drone.

4 Battery Management System

The Battery Management System (BMS) monitors the condition of the battery and helps protect it from unsafe operating conditions.

A simplified BMS structure may include:

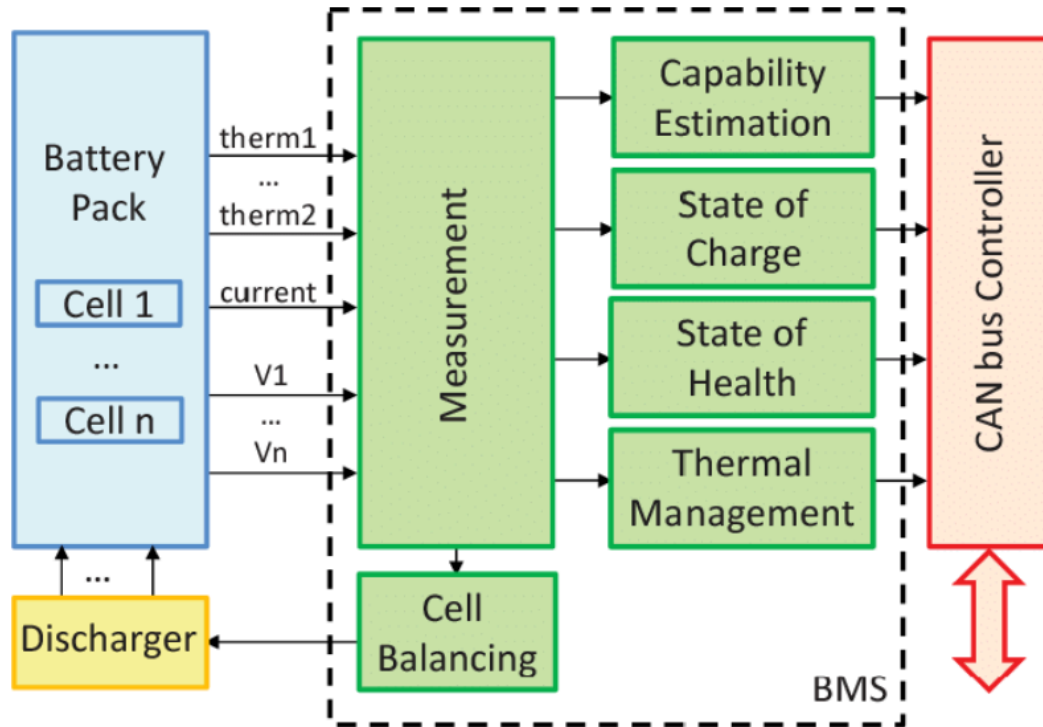


Figure 2: BMS block diagram

5 Power Distribution System

The power is then divided into two main branches:

- **High-Power Branch**
Supplies the propulsion system:
This branch carries the largest current in the drone and therefore requires appropriately rated connectors, wires, PCB copper, and protection devices.
- **Low-Power Branch**
Supplies the control and sensing electronics:

Power Distribution Board → DC-DC Converters → Flight Controller / Sensors / Communication

The battery voltage is converted into regulated voltage rails such as 12 V, 5 V, or 3.3 V, depending on the requirements of the electronics.

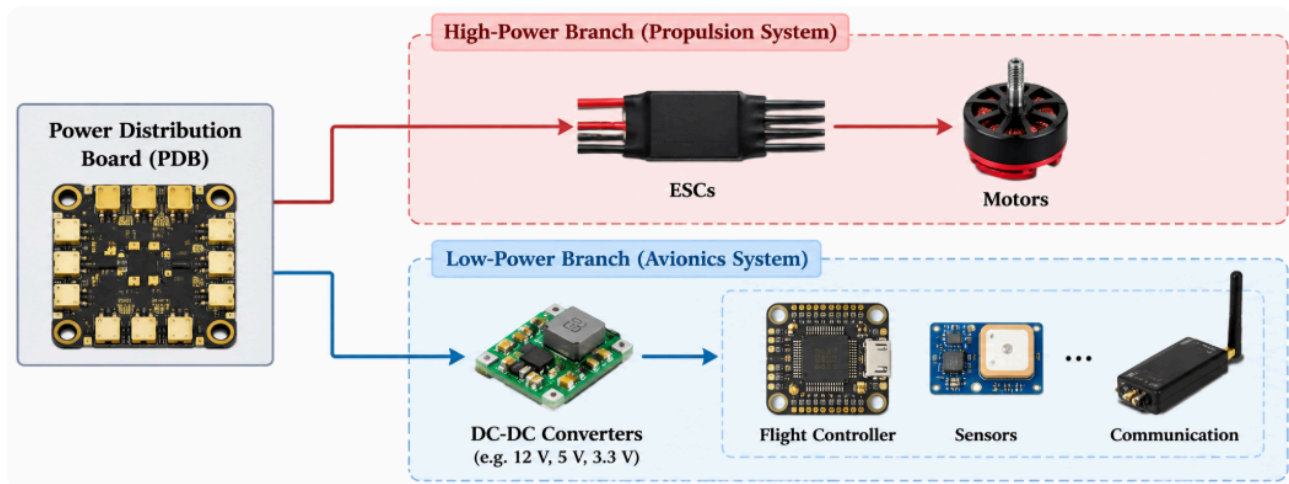


Figure 3: Power Distribution diagram

6 Software

The following tools will likely be useful:

- **MATLAB / Python**

For:

- Battery data analysis
- Flight-time estimation
- Power-budget calculations
- SOC algorithm development

- **LTspice**

For:

Circuit design and analysis

- DC-DC converter simulation
- Voltage measurement circuits
- Current-sensing circuits
- Protection circuitry

- **Cadence**

For:

- Schematic design
- PCB design
- Power distribution board development
- Custom BMS PCB development

- **C / C++**

For implementing battery monitoring and protection logic on the microcontroller.

7 To-Do Over Mid-Semester Break

The goal over the mid-semester is to understand the basic of drone power system and BMS rather than immediately designing a complete BMS and power distribution.

By the end of mid-semester, each member should watch the videos below and have a basic understanding of:

- **Series and parallel battery connections**

► [Series and Parallel Battery Connections](#)

- Battery capacity, energy, and C-rating
 - ▶ [Battery Capacity and Energy](#)
 - ▶ [Battery C-Rating](#)
- What is a Battery Management System (BMS)?
 - ▶ [Introduction to Battery Management Systems](#)
 - ▶ [Battery Management System Overview](#)
- BMS active and passive cell balancing
 - ▶ [Active and Passive Cell Balancing](#)
- Drone power distribution board explained
 - ▶ [Explains Power distribution boards for quadcopter drones](#)

8 Optional Advanced Research

For anyone interested in going further, useful topics include:

- State-of-Charge estimation
- Equivalent circuit battery models
- SOC estimation
- Passive versus active cell balancing
- Power distribution PCB design
- CAN-based battery communication